Anmerkung Conditions Report

Rule engine covering Dachser, K+N, DHL Express, Wackler & Honold · source: assets/anmerkung.js · generated 2026-10-23

How a row is classified

Each worksheet row is read column-wise into the five *process** functions. A row is only classifiable when its **Stat_Freigabe** cell equals **10** (released); otherwise the row is skipped (Wackler still runs its Kontierung check). Each forwarder compares its forwarder-specific **Differenz** columns (FR, MT, SNK, EXP, TZ, …) against a **firing threshold**:

```
hasErr(v, t) = |v| > t
```

A delta that clears the threshold is called a "trigger" and contributes a label. All labels a row gathers are joined with **" // "** into the **Anmerkung** cell (deduplicated by join). Some rules *return early* or block later rules, because a single finding (e.g. a duplicate Fremdnummer booking) makes every other delta on the row spurious.

Thresholds

Constant	Value
Firing threshold — Dachser	|diff| > 0.08 (default 0.08)
Firing threshold — K+N	|diff| > 0.09 (default 0.09)
Firing threshold — DHL Express	|diff| > 0.04 (default 0.04)
Firing threshold — Wackler	|diff| > 0.09 (default 0.09)
Persisted per-user	localStorage key <font face='Courier' size='8'>anme
Kontierung? checks	disabled globally — KONTIERUNG_ENABLED = fal

1. Dachser — processDachser

Gate: Stat = 10. Reads C502/C503 (Einlagern/Auslagern), SNK, ZZ, SAM, DGR, EXP, MT, TZ, C38L, FR with weight/tier logic on the DACHSER_BP rate-card tiers (50 kg steps → 10 000 kg, open 999 999 bucket). Cascade order matters — the first rule that fires colours the rest.

1.0 Blank-Tarif accounting rows (early return)

Condition	Label written to Anmerkung
TARIF cell empty AND |FR| > T; no SACH, no SERV_ART, no other diff (MT/TZ/EXP/SNK/ZZ/DGR/LG/AV/SAM/SBFU/C38L)	`VORHOLUNG`
same empty-TARIF but SACH + SERV_ART are set	`Fremdnummer <row's ReferenzNr> bereits berechnet in RE00123xxx, ok? (falls back to the 5034xxx placeholder when ReferenzNr is empty; the Beleg it was charged in stays a placeholder, it names a document outside this workbook)`
empty TARIF + FR trigger + departure country non-DE	`kein Tarif für <Land>`
empty TARIF + FR trigger + domestic lane above 10 t (Brutto kg > 10000)	`Kein Tarif für DE >10 to, ok?`

1.1 Direct-cost labels (each fires independently)

Condition	Label written to Anmerkung
C196 Differenz over threshold (Kalkulationskosten)	`Kalk. Kosten Zustellung (emitted as the leading segment)`
C502 Kosten DL has a non-zero value	`Einlagern`
C503 Kosten DL has a non-zero value	`Auslagern`
LG Differenz over threshold	`Lagergeld`
AV Differenz over threshold	`Gebühr für vergeblichen Abholversuch`
ZZ Differenz over threshold	`2. Zustellung`
SAM Differenz over threshold	`Samstagzustellung`
DGR Differenz over threshold (skipped when TARIF is zero)	`Gefahrgut-Zuschlag`
EXP Differenz over threshold	`Produktzuschlag, or Terminzuschlag when EXP Kosten DL = 95`
ZABF Differenz over threshold	`Einfuhrzollabfertigung`
SBFU Differenz over threshold	`SBfU-Bescheinigung f. Umsatzsteuerzwecke`
C38L Differenz over threshold	`Maut von/bis NL`

1.2 SNK — switch on billed SNK_DL

Condition	Label written to Anmerkung
SNK_DL = 190 or 95	`AUSFALLFRACHT`
SNK_DL = 130 or 390	`Standgeld`
SNK_DL = 75 + SNK_TARIF > 0 + SNK diff over T	`Differenz Hebebuehnen-Zuschlag`
SNK_DL = 75 + SERV_ART = K1AV	`Speditionskosten gem. Text`
SNK_DL = 75 (no tariff backing)	`Ausfallfracht/Schadensersatz`
SNK_DL = 80 + SERV_ART = K1AV	`Speditionskosten gem. Text // Admin Zeitfensterbuchung Handel`

SNK_DL = 11 (+ diff over T)	`Differenz Telefonische Zustellankündigung - Laderaumzuschlag`
SNK_DL = 14 (+ diff over T, TARIF not zero)	`Differenz Automatische Zustellterminvereinbarung - Laderaumzuschlag (K1AV → Differenz Laderaumkostenentwicklung)`
SNK_DL = 5 + SERV_ART = K1AV (SNK_TARIF=0)	`Admin Zeitfensterbuchung Handel`
SNK_DL = 5 + SERV_ART = K1AV (SNK_TARIF>0)	`Differenz Admin Zeitfensterbuchung Handel - Laderaumzuschlag`
SNK_DL = 5 + diff over T + domestic lane (Abg.-Land blank or DE); skipped when TARIF zero	`Differenz Automatische Zustellterminvereinbarung - Laderaumzuschlag`
SNK_DL = 5 + diff over T + import lane (Abg.-Land not DE, Empf.-Land DE)	`Differenz Laderaumkostenentwicklung`
SNK_DL = 9 (+ diff over T, TARIF not zero)	`Differenz Telefonische Zustellterminvereinbarung - Laderaumzuschlag`
other SNK_DL + SERV_ART = K1AV, or non-integer SNK_DL whose SNK_DIFF > T (a residual equal to the 0.08 threshold does not qualify)	`Differenz Laderaumkostenentwicklung`
K1AV and SNK_DL is a whole 65-multiple (≥ 65) and not one of the Standgeld values (130, 390)	`Wartezeit <n>h á 65 EUR, ok? (the headline alone, no admin-fee suffix)`
K1AV and SNK_DL − 5 ≥ 65, residual a whole 65-multiple	`Wartezeit <n>h á 65 EUR, ok? // Differenz Admin Zeitfensterbuchung Handel - Laderaumzuschlag (the bundled admin fee is still owed, so the phrase is not emitted alone)`
K1AV and SNK_DL − 5 ≥ 65, residual cents-bearing and not a 65-multiple	`<amount> Kosten für 2.Zustellung etc. ok?`
other SNK_DL (+ diff over T)	`Differenz Automatische Zustellterminvereinbarung - Laderaumzuschlag`
Non-integer SNK_DL and SNK_DIFF rounds to 5/9/11/14 (±0.05)	`code re-derived from SNK_DIFF, then the switch above`

1.3 EXP + others

Condition	Label written to Anmerkung
EXP diff over T and EXP_DL = 95	`Terminzuschlag`
EXP diff over T (otherwise)	`Produktzuschlag`
Empf.-Ort = LONDON, PLZ leading letter	`Zone korrekt berechnet? (+ Einfuhrzollabfertigung when EXP clean)`
ZABF Differenz over threshold (Import-Zollabfertigung line)	`Einfuhrzollabfertigung`

1.4 FR — weight/tier decision tree (after SNK)

Condition	Label written to Anmerkung
SERV_ART = K1AU and the rate cards return Sonderfahrt for the lane (FR Kosten DL − 180/150/120/90 lands on a NEW-card cell)	`Sonderfahrt`
Anz. Sdg > 1 (multi-consignment)	`hätte gebündelt werden können?`
REFERENZ3 = ZW (deviating intermediate consignee)	`Differenz aufgrund abweichender Zwischenempfänger <PLZ Ort>`

SERV_ART = K1AS	`Sonderfahrt`
SERV_ART = K1AU and |FR − 90| ≤ 0.1	`"Sonderfahrt" 90 EUR doppelt berechnet?`
weights equal or same tier; FR > 1 on an international lane (non-DE origin or destination)	`Sonderfahrt, or Dachser berechnet die Kosten nach dem bisherigen Tarif — decided by the rate cards: (FR Kosten DL − S) landing on a NEW-card cell for S in 180/150/120/90 gives Sonderfahrt, otherwise FR Kosten lt. Tarif landing on an OLD-card cell (timed by 1.15 on imports) gives the bisherigen wording. Ratecards are encrypted (assets/dachser-ratecards.enc.json); without the passphrase the bisherigen wording stands.`
weights equal or same tier; FR > 1 (or ≥ 0.05 with no other label); ZZ diff = 35	`Fracht Differenz (ZZ=35)`
weights equal or same tier; FR > 1 (or ≥ 0.05 with no other label); otherwise	`Frachtdifferenz`
weights equal or same tier; FR < –1.0	`Differenz aufgrund abweichender Gewichte`
both weights known, different tier, FR 0–1	`Frachtdifferenz`
both weights known, different tier (FR ≥ 1 or negative)	`Differenz aufgrund abweichender Gewichte`
no weight basis, FR negative (or < 1.0)	`Differenz Frachtzu/ abschlag`
no weight basis, FR positive	`Differenz aufgrund von abweichendem Gewicht`
Condition	**Label written to Anmerkung**
SNK_DL = 14 + SNK diff over T (footer)	`Abholterminvereinbarung`
row has no other label, only TZ over T	`Differenz treibstof`

2. K+N — processKN

Gate: Stat = 10. Tiers from KN_BP (like Dachser but collapses the 8500/9500 steps and opens at 99 999). FR (freight) is the lead column; EXP/MT/SNK/TZ add their own labels.

Condition	Label written to Anmerkung
FR over T and ReferenzNr lists multiple docs (contains ',') — early return	`hätte gebündelt werden müssen, ok?`
FR over T, TARIF blank or '-' — flat-rate freight	`Pauschalfracht`
FR over T, Amazon recipient, VKG and VKG_DL in the same weight tier	`hätte nach Amazon Tarif abrechnen müssen`
FR over T, Amazon recipient, VKG and VKG_DL cross tiers	`Differenz aufgrund abweichender Gewichte`
FR > 0 and SNK_DL = 5, same tier	`hätte nach Amazon Tarif abrechnen dürfen`
FR > 0 and SNK_DL = 5, cross tier	`Differenz aufgrund abweichender Gewichte`
any other FR over T	`Differenz aufgrund abweichender Gewichte`
EXP over T	`FIXTERMIN`
MT over T	`Mautdifferenz`

SNK cascade — exact matches first, then tolerance fallback

Condition	Label written to Anmerkung
SNK diff ≈ +9	`Avis, ok?`
SNK diff ≈ –9	`Differenz avis`
|SNK diff| ≈ 25	`Portalavisierung, ok?`
SNK diff over T and SNK_DL = 5 or 25	`Portalavisierung, ok?`
SNK diff over T and SNK_DL = 9	`Avis, ok?`
SNK diff over T and SNK_DL = 12	`B2C-zuschlag, ok?`
SNK diff over T and SNK_DL = 18 (unless Avis already)	`Avis, ok?`
SNK diff over T and SNK_DL = 34	`Portalavisierung, ok? // Avis, ok?`
SNK diff over T, code not matched	`SNK Differenz`
TZ over T AND neither FR nor MT fired	`Differenz treibstoff`
KOST or SACH blank/'–' (disabled)	`Kontierung?`

3. DHL Express — processDHL

Gate: Stat = 10. Uses a **blocking chain**: the first group (EL/FR/PAL/OW/YO/YL/ND/SF + SNK) sets block, and only when nothing blocked does the AC/MT/NX/OS group fire.

Condition	Label written to Anmerkung
TARIF present but value 0 (raw contains '0' or is '-')	`Fremdnummer xxx bereits berechnet in RExxx, ok?.`
EL Differenz over T	`EXTENDED LIABILITY, ok?`
FR Differenz over T	`Differenz aufgrund von abweichendem Gewicht/Volumen`
PAL Differenz over T	`nicht stapelbar ok?`
OW Differenz over T	`Differenz overweight, ok?`
YO positive and divisible by 15	`Non conveyable piece-weight ok?`
YO over T (not a 15-multiple)	`non conveyable piece ok?`
YL over T	`Non-conveyable piece irregular ok?`
ND over T	`Neutral delivery ok?`
SF over T	`Direct signature ok?`
SNK = 5	`DIRECT SIGNATURE , ok?`
SNK = 25	`Limited quantities ok?`
SNK = 30	`Elevated Risk, ok?`
SNK = 60	`Eelevated risk ok? // Restricted destination ok?`
SNK over T, code not matched	`SNK Differenz`
below-block group — AC = 11	`Addres Correction, ok?`
below-block group — AC over T	`Address Correction ok?`
below-block group — MT over T	`Mautdifferenz`
below-block group — NX over T	`demand surcharge ok?`
below-block group — OS over T	`Oversize piece ok?`
no other label, TZ over T only	`Differenz treibstof`

4. Wackler — processWackler

Gate: Stat = 10 (otherwise only the Kontierung check runs). Numbered rules evaluated in order with several early returns. Weight-tier logic uses the Wackler rate-card breakpoints (50 kg steps → 10 000 kg, open 999 999 bucket) shared by the national and international rate sheets.

Terminal (early-return) rules

Condition	Label written to Anmerkung
Empf.-PLZ 88499 and Empf.-Ort RIEDLINGEN with a non-hub Abg.-PLZ (return hub, keyed on the recipient) - first rule, no exceptions: every other phrase is suppressed, since the freight/maut/fuel gaps are return artefacts; a row that starts AND ends at the hub is outbound, not a return	`Return, ok?`
TARIF '-' or '0' (non-empty), or blank + FR/MT/TZ signal — duplicate booking; all delta rules suppressed	`Fremdnummer xxx bereits berechnet in RExxx, ok?`
TARIF blank + SNK signal + no FR/MT/TZ + no known SNK/AVIS code — storage fee	`Lagergeld, ok?`
SNK >= 1.0 × TARIF + no FR/MT/TZ + no codes (placeholder tariff)	`Pauschalfracht, ok?`
|FR Differenz| > T and the MT, SNK, TZ and AVIS Differenz cells all hold exactly zero - the freight is the only cost delta, sign-agnostic; all delta rules suppressed	`Pauschalfracht, ok?`
|FR Differenz| = 250.00 ± 0.5 and the MT, SNK, TZ and AVIS Differenz cells all hold exactly zero - the 250 € surcharge billed on the freight line instead of the SNK line; one finding, both axes, sign-agnostic	`Zusatzkosten 250 €, ok?`

AVIS surcharge codes

Condition	Label written to Anmerkung
AVIS = 7.5 / 8.5 / 6.5 / 8.7 (sign-insensitive)	`Avis, ok?`
AVIS = –8.7 (should have used the cheap telephonic rate)	`hätte Avisgebühr telefonisch abrechnen dürfen`
AVIS = 1 (per-shipment Avisnachweis line)	`Differenz avis, ok?`

SNK surcharge code book (sign-insensitive, tolerance ±0.5)

Condition	Label written to Anmerkung
SNK ≈ 38	`NL-FIX`
SNK ≈ 11.5 (±0.1)	`hätte B2C-Line abrechnen dürfen`
SNK ≈ 22	`2. Zustellung ok?`
SNK ≈ 25	`Terminzustellung`
SNK ≈ 80 on a weighed shipment (weight-gated)	`Terminzustellung`
SNK ≈ 170	`Terminzustellung`
SNK ≈ 180	`Terminzustellung, ok?`
SNK ≈ 43	`2.Zustellung ok?`
SNK ≈ 289	`Umverfügung`
SNK ≈ 49	`NL-12, ok?`
SNK ≈ 113	`NL-SPEZ, ok?`
SNK ≈ 16	`NIGHTFIX`
SNK ≈ 81	`NL-10, ok?`

SNK ≈ 250	`Zusatzkosten 250 €, ok?`

Gewichte / bundling decision tree (FR + both weights real)

Condition	Label written to Anmerkung
multi-ref, weights 1–5% apart (same or cross tier)	`hätte gebündelt werden müssen (alone)`
VKG and VKG_DL in different rate tiers	`Differenz aufgrund abweichender Gewichte`
same tier, ≥ 6 references	`hätte gebündelt werden müssen (alone)`
pallet volume (colli×285) ≥ 10 000 kg or ≥1.05×VKG, tiers apart	`Differenz aufgrund abweichender Gewichte`
same tier, ≥ 3 references (fallback)	`Differenz aufgrund abweichender Gewichte`
otherwise (same-tier rate-card case)	`Wackler rechnet Frachtrate für <tier>kg ab`
cross-tier, multi-ref, VKG_DL ≤ 20% of VKG — TERMINAL	`hätte gebündelt werden müssen (alone)`
cross-tier, multi-ref, FR > 0, weights within 20%	`hätte gebündelt werden müssen (alone)`
cross-tier, far weights or FR credit	`Differenz aufgrund abweichender Gewichte`
same-tier multi-ref with unequal whole-kg weights	`hätte gebündelt werden müssen (alone)`
same-tier multi-ref with (near-)identical weights, non-integer (volumetric) VKG	`Wackler rechnet Frachtrate für <floor tier>kg ab`
no real weights + multi-ref + FR	`hätte gebündelt werden müssen (alone)`

Fallback / additive rules

Condition	Label written to Anmerkung
FR over T, no Gewichte/bundling fired	`Frachtdifferenz`
MT over T	`Mautdifferenz`
|SNK| ≥ 5.0, unrecognised code, not bundled, and no weight/bundling phrase fired	`SNK Differenz`
SNK ≈ –150 (±2)	`hätte Hebebühne abrechnen dürfen`
|TZ| ≥ 2.0, both sides carry a DIFFERENT fuel rate (|TZ_DL/FR_DL – TZ_Tarif/FR_Tarif| > 0.005; a row or layout without both fuel rates keeps the plain threshold), not on same-tier 'Wackler rechnet' row, not an FR credit - 'Differenz treibstoff' when the fuel gap is the row's only finding, 'Differenz Energiezuschlag' when an FR, MT, SNK, AVIS, weight or bundling phrase fired as well; a Hebebühne credit row queries the surcharge instead	`Differenz treibstoff / Differenz Energiezuschlag / Dieselzuschlag ok?`
NL-FIX + FR + blank KOST/SACH — 'Zone korrekt?' prepended, Frachtdifferenz stripped	`Zone korrekt? // …`
only rule left is TZ over T, no other phrase fired	`Differenz treibstoff`
KOST & SACH both blank/X (disabled)	`Kontierung?`

Wackler constants

Constant	Value
|TZ| trigger (rule 10)	2.0
TZ/FR rate band: inside it the fuel delta is derived (rule 10)	0.005
|SNK| noise floor (rule 9)	5.0
SNK ≈ –150 ± 2 → Hebebühne	150 ± 2.0
SNK ≥ ratio × TARIF → Pauschalfracht	× 1.0
SNK ≈ 80 ± 0.5 → Terminzustellung (weighed)	80 ± 0.5

Multi-ref same-tier upper bundle weight	1000
Cross-tier near band (bundling)	0.2
VKG_DL ≤ fraction of VKG → terminal bundling	0.2
Pallet volume estimate per collium (kg)	285
'Near-equal' weights band	0.01
Close-but-unequal bundle band	0.05
Same-tier ≥ N references → bundling	6

5. Honold — processHonold

Gate: Stat = 10, plus the OLD tariff must be unlocked and complete (tiers, zone rates, country codes). The tariff matrix is business data and ships encrypted (assets/honold-tariff.enc.json); until it is decrypted the engine *skips* Honold rows (null) rather than blanking an existing note it cannot re-derive. Two rules, both decided from FR Kosten DL. A ZZ cost line is a second delivery attempt billed on its own: when its Differenz clears the Dachser difference threshold (T_DACHSER, 0.08 by default) and FR Kosten DL is blank / 0, the whole note is "2. Zustellung". A row that carries freight as well as a ZZ charge is decided by the second rule. Otherwise FR Kosten DL is matched against the OLD tariff: the row keeps the previous-tariff note when it equals a zone rate published for its Empf.-Land in the Bis bracket its Volumen kg falls in (every zone of the country is a candidate, because the row carries the country but not the zone, and no OLD amount sits at two different brackets across a country's zones - measured 0 for CH, ES, FR, GB, GR, IT, NO, PL, PT, SE), and reads a weight difference otherwise. Volumen kg DL is not read: it is unreliable when ReferenzNr lists several references.

Condition	Label written to Anmerkung
ZZ Differenz above the Dachser difference threshold (0.08 by default) and FR Kosten DL blank / 0	`2. Zustellung`
FR Kosten DL equals an OLD-tariff rate for the row's Empf.-Land in the bracket its Volumen kg falls in	`Honold berechnet die Kosten nach dem bisherigen Tarif`
FR Kosten DL matches no OLD-tariff rate for the row's Empf.-Land in that bracket (a rate of another bracket, or none at all)	`Differenz aufgrund abweichender Gewichte`
blank / 0 FR Kosten DL, a missing / 0 Volumen kg, or no zone column for the destination country	`Differenz aufgrund abweichender Gewichte`

Honold constants

Constant	Value
Weight brackets	Bis column of the OLD tariff (50 … 10 000, last row c

How the output is consumed

The **Anmerkung** cell is a ' // '-joined list of independent rule outputs. The diff tooling (classifyDiff) compares the engine output against the auditor's ground-truth column after normalising phrase spelling/case/whitespace (normPhrase), and buckets each row as **correct / missed / overfired / wrong**. Phrase-level decomposition then reports which specific phrase the engine missed (*missing_phrases*) and which it over-fired (*extra_phrases*). That comes right from the A/B training pairs — a rule change is only accepted when it re-validates against the git-ignored Soll-Ist workbooks.

Control surface: scripts/anmerkung/verify.mjs smoke-tests that the five process* functions and window.AnmerkungV5 exist; the same periods feed the diagnostics facade. Firing thresholds and the Wackler constants above are read live out of assets/anmerkung.js at build time, so this report tracks the engine — but the cascade descriptions are maintained by hand in scripts/anmerkung/report_conditions.py and should be kept in sync when rules change.